PS-3 Research Guide: Predicting Where an Address Is, and How Sure We Are

Prepared as Principal-Investigator guidance for the "Location Prediction + Confidence Interval" team (CreditNirvana Problem Statement 3: "Address Geocoder That Learns from Field Visits").

> NOTE
>
> Every number in this document comes from actually analysing your CSV files (scripts are in the `scratch/` folder next to this file). Your map uses a **local grid measured in metres** (`x`, `y`), not latitude/longitude. That means a plain ruler-style (Euclidean) distance, `√(Δx² + Δy²)`, gives the error in metres directly.

0. Summary (read this first)

Question	Short answer
Do we need to understand the dataset in depth?	**Yes, definitely.** The data has traps built into it on purpose: fake visits, borrowers met away from home, duplicated files, addresses with no geocode, addresses written in three scripts. A model built without knowing these will learn the wrong locations.
How good is the current (commercial) geocoder?	Median error **376 m**, 90th-percentile error **839 m** against the 100 surveyed addresses.
What is the single most useful signal?	**GPS check-ins from visits where the agent actually reached the door**, *after removing fake visits*. When an address has 2 or more such visits, the median of those check-ins is typically **8–23 m** from the true spot.
Our quick prototype (a few lines of code)	Baseline 376 m → **126 m median** just by (a) using check-ins from all "reached the door" outcomes and (b) removing fraudulent visits.
Where is the remaining error?	**55 of the 100 surveyed addresses have no usable visit at all.** For those, we have to rely on the **address text + landmarks + nearby confirmed addresses**. This is where the next big gain is.
Best overall method	A **Bayesian evidence-fusion model**: start from a *prior* (an initial guess from the text/landmarks/geocoder), then update it with each visit, weighted by how reliable that visit is. Then **calibrate the radius with conformal prediction**, so that "80 % confidence" really means the house is inside the circle 80 % of the time.

Part 1. Dataset Dependency & Feature Selection Strategy

1.1 Do we need to deeply understand the dataset? Yes, for 5 reasons

In many ML tasks you can treat the data as "rows in, prediction out". **This is not one of those tasks.** Here's why:

1. **The ground truth is tiny and hidden.** Only `surveyed_addresses.csv` (100 rows) tells us the *true* location. Everything else is a *clue*. If you don't know which columns are clues and which are truth, you can't build or test anything.
2. **Some clues are lies.** We found one agent (**FA009**) whose check-ins pile up in two 25 m spots (probably home and a tea stall) across **50+ different addresses**, with the **same photo re-used up to 32 times**. If you average those in naively, the model learns that dozens of houses sit in the agent's living room.
3. **The visit "outcome" changes what the GPS means.** "Address not traceable" check-ins are on average **1.6 km** from the true house, while "locked premises" check-ins are only **~20 m** away (the agent stood at the right door, it was just locked). A beginner would throw away "locked premises" as a "failed visit". That would be a mistake.
4. **The structure is relational (several linked tables).** Visits link to addresses, addresses link to accounts, GPS trails link to visits, landmarks link to towns. Most of the useful features only show up when you **join** the tables (see the diagram below).
5. **The problem statement requires it.** It explicitly lists "weak noisy evidence", "fake visits", "one place, many spellings" and "honest confidence" as things you must solve. Each of these is a property of the data.

1.2 Data map: what each file is and how much it matters to us

File	Rows	What it contains	Role for our team	Importance
`ps-3/surveyed addresses.csv`	100	True (surveyed) x, y of an address	**Ground truth / labels** (66 train, 19 val, 15 test)	★★★★★
`shared/field_visits.csv`	5,578	Check-in x, y, GPS accuracy, dwell time, outcome, remark, photo hash	**Main evidence**	★★★★★
`ps-3/visit_gps_points.csv`	160,406	Breadcrumb trail of each visit (~1 point every 30 s)	Trail shape, stop points, fraud detection, "where it's not"	★★★★☆
`ps-3/baseline_geocodes.csv`	2,880	Commercial geocoder pin + precision level	**Prior (starting guess)** and the benchmark we must beat	★★★★☆
`shared/addresses.csv`	3,117	Raw address text, type, source, town	Text → landmark/locality parsing; entity resolution	★★★★☆
`ps-3/landmarks_poi.csv`	240	Known landmarks (temples, ration shops, tanks…) with x, y	Anchors for "near X / behind X"	★★★★☆
`ps-3/localities.csv`	36	Locality name, pincode, centroid	Coarse prior; spelling dictionary	★★★☆☆
`ps-3/towns.csv`	3	Town, address style, radius	Town-specific parsing rules & bounds	★★☆☆☆
`shared/agents.csv`	30	Agent tenure, language team, town	Agent reliability features	★★★☆☆
`shared/splits.csv`	2,400	train / validation / test per account	**Must respect** for honest evaluation	★★★★☆
`shared/accounts.csv`, `lenders.csv`	2,400 / 6	Loan & lender info	Minor (lender → address format style)	★☆☆☆☆
`shared/dial_attempts.csv`, `payments.csv`	51k / 2k	Phone calls, payments	**Mostly out of scope** (that's PS-1 / PS-2)	☆☆☆☆☆

WARNING

Data hygiene issues to mention in the report:

- `field_visits.csv` and `field_visits (1).csv` are **byte-for-byte identical**. Use only one, or every visit gets counted twice.

- **237 addresses (town `OUT`, type `permanent_native`, e.g. "Village Devipur, Taluk A")** have **no baseline geocode** and no landmarks. Give them a very wide confidence radius or mark them "cannot geocode", and report them separately.
- `salary_credit_day` and `ability_to_pay_estimate` contain many missing values, but they don't matter for location.

1.3 What the data told us (key evidence)

(a) The commercial geocoder's error depends heavily on its `precision` label

Precision	n (surveyed)	Median error	90th percentile
rooftop	1	26 m	n/a
street	16	109 m	166 m
locality	73	386 m	627 m
pincode	10	1,376 m	3,820 m

→ `precision` is a **direct predictor of uncertainty**. It is the first feature for the confidence radius. About 71 % of all addresses are geocoded only at `locality` level.

(b) Check-in distance to truth depends heavily on the visit outcome

Outcome	Median distance to truth	What it means
met_family	16 m	Agent was at the house ✅
locked_premises	20 m	Agent was at the house (it was locked) ✅
neighbour_says_shifted	20 m	Agent was at the house (borrower moved) ✅
met_borrower	34 m	Usually at the house; sometimes met at shop or road ⚠️
cash_collected / no_such_person	~70 m	Small sample
address_not_traceable	**1,603 m**	Agent was searching in the wrong place ❌

IMPORTANT

> **Insight #1:** "Did the agent reach the door?" matters more than "Was the visit successful for collections?". `locked_premises` and `neighbour_says_shifted` are *collections failures* but *location successes*. Using them increased coverage from 34 → 45 of the 100 surveyed addresses and reduced median error from 161 m → 126 m.

(c) Repeated, independent visits help a lot

# of door-reached visits for the address	Median error	90th percentile
1	40 m	**5,060 m** (one bad visit wrecks it)
2	14 m	156 m
3	8 m	69 m
4+	23 m	45 m

→ With one visit you can't tell whether it is an outlier. With 2 or more you can. **The number of independent visits (and agents) is a key feature for the confidence radius.**

(d) GPS accuracy matters; dwell time barely does

- Check-ins with `gps_accuracy_m` ≤ 7.7 m: median error **13 m**. With accuracy > 15.7 m: **53 m**. → Use accuracy as a weight.
- Dwell time showed **no clear trend** (24–35 m across all quartiles). It is useful for *fraud* detection but not as a location weight. (Good to report honestly: not every intuitive feature works.)

(e) The fraud pattern (agent FA009)

Signal	Honest visits	FA009's fake visits
GPS trail points	~26	4–9
Trail path length	~1,000 m	60–400 m
Distance between trail start and end	~400 m	3–26 m (didn't move)
Photo hash	unique	re-used (`ph0080` – `ph0085`), up to 32×
Check-in location	spread over town	two 25 m "hotspot" cells covering 50+ addresses
Remark vs outcome	consistent	contradictory ("met_borrower" + remark "ghar band tha" = house was locked)

(f) "Not traceable" visits are negative evidence

On "not traceable" visits the agent's check-in was **230 m from the commercial pin** but **681 m from the truth** (surveyed addresses, honest agents). In other words, agents search around the wrong pin. Their GPS trail tells us **where the house is NOT**, which is useful to push the estimate away from those areas.

(g) One building, many spellings

We found **62 pairs of addresses from different accounts whose confirmed check-ins are within 15 m** of each other, e.g.:

- `No. 108, Gali no-3, Ward 11, near Masjid, Patel Ngr…` ⟷ `h.no. 243 गली नं. 3 ward 11 patel nagar…`
- `c block 4th rd nr. post office silver oak layt…` ⟷ `#181 c block 4th road jama masjid ke paas silver oak lyt…`

→ When we resolve these as "same place / same street", **one confirmed visit can improve many addresses**.

(h) Address text is messy but structured

- 8.4 % contain non-Latin script (Kannada ಚಚ್೯ ಹತ್ತಿರ, Hindi गली नं. 3).
- Typos and abbreviations: `Layt`, `Lyt`, `Colny`, `Nagr`, `Ngr`, `Hoase`, `bhd`, `nr.`, `opp`, `4th X` (X = cross).
- Town-specific grammar: **T1 Kaveripura** (Karnataka style: *Cross / Main / hattira*), **T2 Devgarh Nagar** (Hindi style: *Gali no. / Ward / ke paas*), **T3 Navanagara East** (metro style: *Block / Road*).
- Agent remarks contain directions: *"nija mane Milk Dairy hinde ide, 2 cross munde"* (Kannada: "the real house is behind Milk Dairy, 2 crosses ahead"), *"asli ghar Sarkari School ke peeche hai, 2 gali aage"* (Hindi: "the real house is behind the government school, 2 lanes ahead").
- Surveyed true locations are a median of **160 m** from the nearest known landmark (max 438 m). So a landmark mention alone narrows things down to a few hundred metres.

1.4 Answer for the senior leads: "What features could we better use to predict the location well?"

We group the features into six families, ranked by expected impact.

Family A: Visit evidence features (highest impact)

These describe each individual "clue" (one visit).

Feature	How to compute	Why it helps
`door_reached`	1 if outcome ∈ {met_borrower, met_family, cash_collected, locked_premises, neighbour_says_shifted, no_such_person}, 0 if `address_not_traceable`	Separates "agent stood at the house" (~20 m) from "agent was lost" (~1.6 km).
`outcome_type` (categorical)	Raw outcome	Each outcome has its own error distribution, so the model learns a separate weight for each.
`gps_accuracy_m`	Raw	Smaller value = more precise fix. Becomes the noise σ in the likelihood.
`stay_point_xy`	From the trail: the centroid of the longest period where the agent stayed within ~30 m (a "stay point")	Often closer to the door than the tap-to-check-in point. A good second estimate.
`trail_end_xy`	Last GPS point	Median 11–18 m from truth for door-reached visits.
`checkin vs trail_end_dist`	Distance between check-in and trail end	Large gaps suggest GPS jumps or tampering.
`met away_from_home`	Remark contains *dukaan / shop / angadi / office / kelsakke / kaam*	Borrower may have been met at work. Down-weight for *residence* addresses, up-weight for *office* addresses.
`address_type`	residence / office / permanent_native	Office addresses + "met at shop" = consistent evidence.

Family B: Integrity / trust features (protects against poisoning)

These answer: "Should we believe this clue at all?"

Feature	How to compute	Why it helps
`photo_hash_reuse_count`	How many visits share this photo hash	Real photos are unique. Re-use is a strong sign of fraud.
`agent_hotspot_score`	For each agent, count how many **distinct addresses** were checked in from the same 25 m cell	Real agents don't "visit" 50 different houses from one spot.
`trail_n_points`, `trail_path_len`, `trail_displacement`	Trail statistics	Fake visits have a short trail with almost no movement.
`trail_tortuosity`	path length ÷ displacement	High = searching. Also very high on fake "jitter in place" trails.
`remark_outcome_contradiction`	NLP check: "met" outcome but remark says "locked / not found"	Copy-paste or false logging.
`agent_consensus_residual`	How far this agent's check-in is from the consensus of *other* agents at the same address	Robust per-agent reliability score. Also lets you **cap any single agent's influence**, which the problem statement requires.
`agent_tenure_months`	From `agents.csv`	New agents may be noisier (a hypothesis to test).

Family C: Address-text features (needed for the 55 % with no visit)

Feature	How to compute	Why it helps
`locality_id` (resolved)	Fuzzy-match text to `localities.csv` after normalising spelling (`Colny→Colony` , `Ngr→Nagar`) and transliterating	Gives a locality centroid prior that is more reliable than a pincode.
`pincode valid` / `pincode locality_mismatch`	Compare the 6-digit pin with the locality's pin (e.g. `98304` is a typo)	Detects bad geocoder inputs, so widen the radius.
`landmark_mentions` (list of type + relation)	Multilingual extraction: "behind Water Tank", "ಮಸೀದಿ ಹತ್ತಿರ" (near masjid), "बारात घर के पास" (near the marriage hall) → (type=water_tank, relation=behind)	Each landmark is an *anchor* with known x, y. The house is a few hundred metres from it at most.
`landmark_candidates_xy`	All POIs of that type within the resolved locality	Usually one or two candidates, which turns the problem into a small search.
`street_tokens`	Parse *Nth Cross / Nth Main / Gali no. N / Block X / Road N*	Same street across accounts lets confirmed neighbours constrain the location.
`text completeness_score`	Has house no.? street? landmark? locality? pin?	Incomplete text → wider radius. Also a PS-2 "fabricated at origination" signal.
`script mix` , `lender_kyc_format`	Share of non-Latin characters; lender's KYC format (`kyc_upper` , `app_free_text` …)	Predicts how noisy the text is, which feeds uncertainty.

Family D: Remark-derived correction features (novel)

Feature	How to compute	Why it helps
`remark_correction_landmark` + `relation` + `steps`	Parse *"nija mane Milk Dairy hinde ide, 2 cross munde"* → (milk_dairy, behind, +2 crosses)	Turns free text into a *new spatial clue* that points to an area.
`remark_correction_count`	How many visits repeat the same correction	Agreement between agents = reliability.

TIP

Present Family D as a **hypothesis to test**: in our quick check, visits with correction remarks were already close to the truth (13–39 m). So the main value is probably for addresses where the *pin* is wrong but an agent left a correction. Test it on the validation split before claiming a gain.

Family E: Spatial-context / cross-account features (novel)

Feature	How to compute	Why it helps
`entity_cluster_id`	Entity resolution: group addresses that are the same building/street (text similarity + shared landmark + nearby confirmed check-ins)	**One confirmed visit improves many addresses.** This is explicitly asked for.
`neighbour_confirmed_offset`	For confirmed addresses in the same locality/street, the average (truth – geocoder) shift	Commercial geocoders make *systematic* errors in an area (e.g. every pin in Locality X is shifted 300 m north). Learn the shift and correct it.
`n_confirmed_within_200m`	Count of confirmed locations near the current estimate	Dense evidence nearby = narrower radius.

Family F: Prior features (the "starting guess")

Feature	Why
`geocoder_xy`, `geocoder_precision`	The fallback location and a strong uncertainty predictor.
`locality_centroid_xy`, distance(geocoder, locality centroid)	If the geocoder pin is far from the locality centroid, one of them is wrong.
`town_radius_m`	Hard boundary: an estimate outside the town is impossible.
`negative_search_mask`	Area covered by "not traceable" trails

What NOT to use (and why that matters)

- **Loan and financial fields** (`outstanding`, `bureau_score_band`, `dpd`, …) and **dial/payment logs**: these have no causal link to where a house is, so they only add noise. Using them would also go against the **DPDP Act's purpose limitation**: "the system does not infer locations beyond what collections needs."
- **Outcome as a model input when predicting a *new* address with no visits**: it's only available after a visit. Be careful about *leakage* (accidentally using information that would not exist at prediction time).

1.5 Why this feature focus maximises accuracy

Think of each address as a **detective case**:

- The **prior** (geocoder, text, landmarks) is the initial theory: "probably somewhere in this neighbourhood."
- Each **door-reached visit** is an eyewitness. Two consistent eyewitnesses beat one. A witness with a blurry view (poor GPS accuracy) counts less.
- **Integrity features** identify witnesses who are lying (fake visits), so they get ignored.
- **Text, landmark and neighbour features** let us solve cases that have *no* eyewitness yet. That is where 55 % of the error is now.
- Our prototype already shows the direction: **376 m → 126 m median** with just two of these ideas. The remaining 90th-percentile error (~620 m) comes almost entirely from **unvisited addresses**, so Families C, E and F are where the next gain will come from.

Part 2. Location Prediction & Confidence Theory

2.1 How to frame the problem: "a probability cloud, not a single dot"

A beginner's instinct is to predict one dot (x, y). A better approach is to predict a **probability distribution**: a "cloud" that is dense where the house probably is and thin where it probably isn't. From this cloud we get:

- **The pin**: the densest point (or the centre of the cloud).
- **The confidence radius**: the circle around the pin that contains, say, 80 % of the cloud.

This approach is called **Bayesian inference**:

$$\underbrace{p(\text{location} \mid \text{evidence})}_{\text{posterior: our final belief}} \propto \underbrace{p(\text{location})}_{\text{prior: text, geocoder}} \times \prod_{i=1}^{N} \underbrace{p(\text{visit}_i \mid \text{location})}_{\text{likelihood of each clue}}$$

```
posterior: our final belief
p(location | evidence) ∝ prior: text, geocoder
p(location) × i=1∏N likelihood of each clue
p(visiti | location)
```

In plain words: *final belief = initial guess × how well each clue fits*. Every new visit multiplies in a new factor, so the model **"keeps improving as new visits come in"**, which is exactly what the problem statement asks for.

2.2 The building blocks (models and maths), from simple to advanced

Level 1: Robust averaging (start here)

- **Mean vs median:** The ordinary average (mean) is pulled by one crazy point (a fake visit 7 km away). The **geometric median** (the 2-D point that minimises the *total distance* to all check-ins) is robust: it ignores outliers until they make up about half of the data. Computed with the **Weiszfeld algorithm** (a simple loop).
- **Weighted version:** weight each visit by $w_i = \frac{\text{trust}_i}{\sigma_i^2}$wi = σi2trusti, where σ_i σi is GPS accuracy (plus an outcome-specific extra error) and trust_i trusti comes from the fraud model. This is called **inverse-variance weighting**: precise clues count more.
- **Clustering for multiple modes:** if visits form two groups (home vs shop), **DBSCAN / HDBSCAN** finds the groups. Pick the cluster supported by the most independent agents.
- **RANSAC** ("random sample consensus"): repeatedly pick a few points, see how many others agree, keep the best consensus. A classic way to throw out outliers.

Level 2: Probabilistic measurement model (the core of the report)

Each check-in z_i zi is modelled as the true location μμ plus noise. Because some visits are "met on the road" or fake, use a **mixture model**:

$$p(z_i \mid \mu) = \pi_i \cdot N(z_i;\ \mu,\ \sigma_i^2 I) + (1 - \pi_i) \cdot \text{Outlier}(z_i)$$

```
p(zi | μ) = πi · N(zi; μ, σi2I) + (1 − πi) · Outlier(zi)
```

- NN is a Gaussian ("bell curve") around the true spot: the visit was at the house, with GPS noise σ_i σi.
- "Outlier" is a broad distribution (uniform over the town, or a wide Student-t): the visit was somewhere else.
- π_i πi is the **probability this visit is genuine at the house**, *predicted from features* (outcome, fraud score, remark, address type) by a small classifier such as logistic regression or gradient boosting.
- Solve with the **EM (Expectation–Maximisation) algorithm**: alternately (E) guess which visits are outliers, then (M) re-estimate the location from the inliers. It is simple and explainable ("we ignored visit VS001522 because it had a 92 % outlier probability").

Using a **Student-t** distribution instead of a Gaussian has a similar effect, since it has "heavy tails" so outliers don't drag the estimate. See **Huber / M-estimators** for the theory of robust estimation.

Level 3: Grid / particle posterior (handles everything at once)

Divide the town into a grid of, e.g., 10 m × 10 m cells. For each cell compute

log-prior(text, landmark, geocoder, neighbours) + Σ log-likelihood(visits) − penalty(searched by "not traceable" trails).

This is a **grid-based Bayes filter** (the robotics term is *occupancy / localisation grid*; the version with random samples is a **particle filter**).

- It naturally handles **multi-modal** beliefs ("either behind the temple or behind the school").
- It's the cleanest way to add **negative evidence** ("an agent walked here and didn't find it").
- The confidence radius is read directly off the grid: the smallest circle around the pin that contains 80 % of the probability.
- It works **offline** on a phone (a constraint in the problem statement), since the grid for one address is tiny.

Level 4: Learned models (machine learning on top)

These use the 66 + 19 surveyed labels (plus pseudo-labels, see §2.4) to *learn* weights instead of setting them by hand:

Model	What it does here	Why it's useful
Gradient boosting (LightGBM / XGBoost / CatBoost)	Predicts (i) π_iπi, the probability a visit is "at the door"; (ii) the expected error of the final estimate	Works very well on small tabular data and is explainable with **SHAP values** (regulators need this).
Residual / offset regression	Predicts the correction vector (Δx, Δy) from the geocoder pin using locality, precision and neighbour offsets	Learns systematic geocoder bias per area.
Gaussian Process regression (kriging)	Smoothly interpolates the geocoder error field across space from confirmed addresses	Gives a mean *and* a variance, so you get uncertainty for free. Comes from geostatistics.
Quantile regression (e.g. LightGBM with `objective='quantile'`)	Directly predicts the 80th/90th percentile of the error distance	A simple, direct way to produce a radius.
NGBoost / Mixture Density Networks	Predict a full probability distribution instead of one number	Advanced option for a "distributional" model.
Deep learning	Mainly for **text understanding**: multilingual transformers (MuRIL, IndicBERT, XLM-R) for landmark/locality extraction; optionally **Graph Neural Networks** over the address–landmark–visit graph	With ~3 k addresses and 100 labels, *training* a deep location regressor from scratch would overfit. Use pretrained text models and keep the location fusion probabilistic. **Say this explicitly in the report**: choosing *not* to over-use deep learning is a sound decision.

NOTE

About "trilateration": trilateration means locating a point from its *distances* to known points (like GPS satellites or cell towers). We don't have distance measurements here; we have noisy *position* readings. The right family of methods is **estimation / sensor fusion** (Levels 1–3). You can mention trilateration in the literature review as related work and explain why it doesn't apply. Reviewers like that kind of reasoning.

Text → location sub-pipeline (for addresses with no visits)

1. **Normalise:** lowercase; expand abbreviations (`nr→near`, `opp→opposite`, `bhd→behind`, `X→cross`, `Rd→road`, `Blk→block`); fix typos with a locality dictionary and **edit distance / Jaro-Winkler / character n-gram TF-IDF**.
2. **Transliterate:** Kannada/Devanagari to Latin (e.g. `indic-transliteration`, AI4Bharat **IndicXlit**), so "ಮಸೀದಿ", "मस्जिद", "masjid" become one token.

3. **Extract entities:** house no., street (cross/main/gali/block/road), landmark + spatial relation (near / behind / opposite / *ke paas* / *ke peeche* / *hattira* / *hinde*), locality, pincode. Start with rules + gazetteer (fast, explainable). Upgrade to fine-tuned **MuRIL/IndicBERT token classification (NER)** if time permits.
4. **Resolve:** match to `localities.csv` and `landmarks_poi.csv` within the town. This is called **toponym resolution / geoparsing**.
5. **Spatial relation model:** "near X" means a distance distribution around X (fit it from confirmed addresses; data says median ~160 m). "behind / opposite" can add a direction if you have street geometry, otherwise treat them like "near".
6. **Combine** with the geocoder pin and the locality centroid into the prior (Level 3 grid).

Entity resolution ("one place, many spellings")

- **Blocking:** only compare addresses in the same town + locality + street token (keeps it fast).
- **Pairwise similarity:** text similarity, shared landmark, shared house number, distance between visit clusters.
- **Decision:** the classic **Fellegi–Sunter** probabilistic record linkage, or a simple classifier, then **connected components / graph clustering** to form "place IDs".
- **Benefit:** pooling visits across accounts and lenders gives more eyewitnesses per place.

2.3 Confidence interval theory: "honest confidence"

What exactly is the "confidence radius"?

For a target level αα (e.g. 0.8), the radius r_α rα should satisfy

$$P(\,\|\text{true} - \text{pin}\| \le r_\alpha\,) = \alpha.$$

P (‖true – pin‖ ≤ rα) = α.

In plain words: *"If we draw an 80 % circle on 100 different addresses, about 80 of the houses should actually be inside their circles."* That property is called **calibration**. Too many inside = circles too big (wasteful). Too few = overconfident (dangerous, since agents trust bad pins). The problem statement warns: *"an uncalibrated score is worse than none."*

Ways to get the radius (from basic to best)

1. **From the Gaussian posterior (analytic).** If the posterior is a 2-D Gaussian with equal spread σ in x and y, the distance error follows a **Rayleigh distribution**, so
   $r_\alpha = \sigma\sqrt{-2\ln(1-\alpha)}$ $\quad$ (e.g. $r_{0.8} \approx 1.79\sigma$, $r_{0.9} \approx 2.15\sigma$).rα = σ −2 ln(1 – α)

   (e.g. r0.8 ≈ 1.79 σ, r0.9 ≈ 2.15 σ). With unequal spread you get an **error ellipse** using the **chi-square distribution** with 2 degrees of freedom. A related term is **CEP (Circular Error Probable)**, the 50 % radius used in navigation and military targeting.

2. **From the grid/particle posterior.** Take the smallest circle containing α of the probability mass (a **highest posterior density region**).
3. **Quantile regression on the error.** Train a model to predict the α-quantile of the distance error from features (number of visits, precision, fraud share, landmark found?…).
4. **Conformal prediction (recommended final layer).** Take *any* of the above raw radii and **calibrate it on held-out data**:
    - Compute a "non-conformity score" on calibration addresses: $s_j = \frac{d_j}{\hat{r}_j}$sj = r^jdj (true error ÷ predicted raw radius).
    - Take the $\lceil (n+1)\alpha \rceil$⌈(n + 1)α⌉-th smallest score, qq.
    - Final radius for a new address: $r = q \cdot \hat{r}$r = q · r^.
    - **Guarantee:** under the mild assumption that calibration and future addresses are *exchangeable* (drawn from the same mix), the coverage is **at least α**, whatever the underlying model. That is the "accuracy proof" your report needs.
    - **Mondrian (group-conditional) conformal:** calibrate *separately* per group (e.g. "has ≥ 2 visits" vs "text only", or per town) so coverage holds in every group, not only on average.
    - **Conformalized Quantile Regression (CQR)** combines approach 3 with conformal calibration and is a strong, popular choice.

How to *prove* the confidence is honest (evaluation)

- **Coverage table:** for α = 0.5, 0.8, 0.9, report the share of test addresses inside their circles. It should be ≈ α.
- **Reliability diagram / calibration curve:** plot nominal α (x-axis) vs observed coverage (y-axis). A perfect model lies on the diagonal.
- **Sharpness:** median radius. Among calibrated models, *smaller circles are better*.
- **Proper scoring rules** (e.g. log-likelihood of the truth under the posterior, or the **CRPS / energy score**). These reward being both accurate and honest.
- **Bootstrap confidence intervals** on your metrics: with only ~15–19 test addresses, report e.g. "median error 45 m (95 % CI 30–70 m)". This is a sign of statistical maturity.

2.4 Validation strategy (critical with only 100 labels)

1. **Respect `splits.csv`:** tune on train + validation and report the final numbers **once** on test.
2. **Pseudo-ground-truth to get more labels:** for addresses with ≥ 3 honest door-reached visits from ≥ 2 agents, **hold out one visit** and use the consensus of the others as "approximate truth" (**leave-one-visit-out**). This turns hundreds of addresses into extra (noisier) labels. Clearly label them as pseudo-labels in the report.
3. **Simulated cold start:** take well-visited addresses, *hide* all their visits, and test the text-only pipeline against the surveyed truth. This measures performance on the 55 % with no visits.

4. **Incremental learning curve:** plot error vs number of visits used (0, 1, 2, 3…). It directly shows "learns from field visits".
5. **Adversarial test:** inject synthetic fake visits (an agent's home location) and show the estimate barely moves. This demonstrates "no single agent can push the model off course."
6. **Spatial leakage check:** addresses in the same entity cluster must not be split across train and test, or you'll overestimate accuracy.
7. **Metrics required by the problem statement:** median and **P90 distance error vs the commercial geocoder**, plus **radius calibration**.

2.5 Recommended final system (what to actually build and defend)

Stage	Method	Explainability hook
1. Clean	Deduplicate files, normalise text, flag `OUT` addresses	Data quality appendix
2. Trust	Gradient-boosted fraud/integrity score per visit + per-agent cap	SHAP: "photo reused 32×, trail moved 9 m"
3. Prior	Text parse → landmark/locality anchors + geocoder (precision-dependent σ) + neighbour offset	"Pin based on: near Water Tank, Basava Nagar"
4. Fuse	Grid Bayes posterior with outlier-mixture likelihood (EM / grid)	"3 visits used, 1 down-weighted (met at shop)"
5. Calibrate	Mondrian split-conformal on the radius	Coverage guarantee
6. Output	Pin, 80 % radius, nearest 2 landmarks + direction ("~120 m NE of Hanuman Temple")	Agent-friendly directions
7. Update	Each new visit multiplies into the posterior; cluster-mates update too	Learning curve plot

2.6 Keywords and literature to back the approach

Search keywords (Google Scholar / Semantic Scholar / arXiv):

- *Geocoding & address understanding:* "geocoding accuracy positional error", "address parsing deep learning", "geoparsing toponym resolution", "Indian address parsing", "multilingual named entity recognition Indic", "transliteration Indic languages", "gazetteer matching", "landmark-based address", "spatial relation extraction natural language".
- *Fusion & estimation:* "Bayesian sensor fusion", "robust estimation outliers GPS", "geometric median Weiszfeld", "M-estimator Huber", "EM algorithm Gaussian mixture outlier", "Student-t robust Kalman filter", "particle filter localization", "grid-based Bayes filter", "inverse variance weighting".

- *GPS trajectories:* "stay point detection GPS trajectory", "trajectory data mining", "GPS spoofing / fake check-in detection", "crowdsourced location truth discovery", "truth discovery source reliability".
- *Spatial statistics:* "kriging Gaussian process spatial interpolation", "spatial autocorrelation", "geographically weighted regression".
- *Entity resolution:* "record linkage Fellegi-Sunter", "entity resolution blocking", "address matching fuzzy string".
- *Uncertainty & calibration:* "conformal prediction", "conformalized quantile regression", "Mondrian conformal prediction", "probabilistic forecasting calibration sharpness", "proper scoring rules", "circular error probable", "Rayleigh distribution radial error", "uncertainty quantification regression".

Foundational references (well-established; verify exact details when you write the bibliography):

Topic	Reference
Geocoding quality	Goldberg, Wilson & Knoblock (2007), *From Text to Geographic Coordinates: The Current State of Geocoding*, URISA Journal; Zandbergen (2008), *A comparison of address point, parcel and street geocoding techniques*, Computers, Environment and Urban Systems; Melo & Martins (2017), *Automated geocoding of textual documents: A survey of current approaches*, Transactions in GIS
Probabilistic localisation	Thrun, Burgard & Fox (2005), *Probabilistic Robotics* (Bayes filters, particle filters, grid localisation); Kalman (1960)
Robust estimation	Huber (1964), *Robust Estimation of a Location Parameter*; Weiszfeld (1937) geometric median; Fischler & Bolles (1981), RANSAC
EM / mixtures	Dempster, Laird & Rubin (1977), *Maximum Likelihood from Incomplete Data via the EM Algorithm*; Bishop (2006), *Pattern Recognition and Machine Learning*
Clustering	Ester et al. (1996), DBSCAN; Campello, Moulavi & Sander (2013), HDBSCAN
GPS trajectories	Li et al. (2008), *Mining user similarity based on location history* (stay points); Zheng et al. (2009), *Mining interesting locations and travel sequences from GPS trajectories*; Zheng (2015), *Trajectory Data Mining: An Overview*, ACM TIST
Truth discovery	Li et al. (2016), *A Survey on Truth Discovery*, SIGKDD Explorations
Spatial statistics	Rasmussen & Williams (2006), *Gaussian Processes for Machine Learning*; Cressie (1993), *Statistics for Spatial Data*
Record linkage	Fellegi & Sunter (1969), *A Theory for Record Linkage*, JASA; Christen (2012), *Data Matching*
Indic NLP	Khanuja et al. (2021), *MuRIL: Multilingual Representations for Indian Languages*; Kakwani et al. (2020), *IndicNLPSuite* (IndicBERT); Madhani et al. (2022/23), *Aksharantar / IndicXlit* transliteration
Tabular ML & explainability	Ke et al. (2017), LightGBM; Lundberg & Lee (2017), SHAP
Uncertainty	Vovk, Gammerman & Shafer (2005), *Algorithmic Learning in a Random World*; Romano, Patterson & Candès (2019), *Conformalized Quantile Regression*; Angelopoulos & Bates (2021/23), *A Gentle Introduction to Conformal Prediction*; Gneiting & Raftery (2007), *Strictly Proper Scoring Rules*; Duan et al. (2020), NGBoost; Lakshminarayanan et al. (2017), Deep Ensembles; Kendall & Gal (2017), *What Uncertainties Do We Need…*; Guo et al. (2017), *On Calibration of Modern Neural Networks*

CAUTION

Before you cite anything, open the paper and check the authors, year and venue yourself.

AI-generated reference lists can contain small mistakes, and a wrong citation hurts credibility.

Part 3. Final Report Architecture

Suggested title: **"Learning Where Addresses Are: Reliability-Weighted Bayesian Fusion of Field-Visit Evidence and Multilingual Address Text with Calibrated Confidence Radii"**

#	Section	What goes in it	Key figures / tables
0	**Abstract** (≈200 words)	Problem, approach, headline result (e.g. "median error 376 m → X m; 80 % radius achieves 0.8x coverage")	n/a
1	**Introduction**	Why field collections fail on vague Indian addresses; cost per visit (₹150–400); scope (location only, *not* whether the borrower still lives there, which is PS-2); our contributions as 3–4 bullets	Example address + map showing geocoder pin vs true house
2	**Problem Formulation**	Formal definition: input (text, visits, trails), output (pin, r_α rα); notation; metrics (median/P90 error, coverage, sharpness); constraints (DPDP, explainability, offline, anti-gaming)	Notation table
3	**Data Understanding**	The data map (§1.2), ER diagram, data quality issues (duplicate file, `OUT` addresses), the error analyses in §1.3	Error by precision; error by outcome; visits-count curve; fraud hotspot map
4	**Literature Review**	Geocoding, robust estimation, Bayesian fusion, trajectory mining, Indic NLP, entity resolution, conformal prediction; why trilateration and pure deep learning don't fit here	Comparison table of approaches
5	**Feature Engineering**	Families A–F with definitions, reasons, and the **ablation** (gain from each family)	Feature table; ablation bar chart; SHAP summary
6	**Methodology**	6.1 Text pipeline; 6.2 Integrity model; 6.3 Prior construction; 6.4 Likelihood/outlier-mixture & Bayesian fusion (equations); 6.5 Entity resolution; 6.6 Confidence radius & conformal calibration; 6.7 Online update	Architecture flowchart; worked example for one address
7	**Experimental Setup**	Splits, pseudo-labels, cold-start simulation, adversarial test, baselines (commercial geocoder, locality centroid, naive mean of check-ins, geometric median), hyperparameters	Baseline list
8	**Results**	Main table (median, P90, coverage@80/90, median radius) vs baselines, with bootstrap CIs; results by town / precision / #visits; learning curve; calibration plot	Main results table; reliability diagram; CDF of errors
9	**Accuracy & Calibration Proofs**	Conformal coverage guarantee (statement + short proof sketch); empirical coverage; robustness to fake visits (adversarial plot); per-agent influence bound	Coverage table; "estimate drift vs # injected fakes" plot
10	**Error Analysis**	Top 10 worst cases and why (no visits, wrong landmark, met at shop…); what would fix them	Case-study maps
11	**Integration, Ethics & Compliance**	How the output feeds the field app, visit planner and PS-2 (location confidence); DPDP purpose limitation; agent-monitoring safeguards; explainability for auditors; offline packaging	Integration diagram

#	Section	What goes in it	Key figures / tables
12	**Limitations & Future Work**	100 labels only; synthetic local grid; street geometry not available; future: OSM road network, active "verify-first" visits, GNN over the address graph	n/a
13	**Conclusion**	1 paragraph	n/a
A	**Appendices**	Glossary, full feature dictionary, abbreviation/transliteration dictionary, hyperparameters, code repository layout, reproducibility checklist	n/a

TIP

Writing tip: in Section 6, walk through **one real address end-to-end** (e.g. an address with a fake FA009 visit plus two honest visits). Show the prior, each visit's weight, the posterior, and the final circle. A worked example convinces readers far more than equations alone.

Appendix: Glossary for ML beginners

Term	Plain-English meaning
Ground truth / label	The correct answer we compare against (here: the surveyed x, y).
Feature	An input measurement given to a model (e.g. GPS accuracy, outcome type).
Feature engineering	Creating new, more useful inputs from raw data (e.g. "trail displacement" from raw GPS points).
Prior / Likelihood / Posterior	Belief *before* evidence / how well evidence fits a hypothesis / belief *after* combining both.
Outlier	A data point far from the rest, often wrong (fake visit, met elsewhere).
Robust estimator	A method that isn't fooled by a few outliers (e.g. median instead of mean).
Gaussian (normal) distribution	The bell curve; describes typical random noise like GPS error.
EM algorithm	A two-step loop: guess hidden labels (outlier or not), re-fit, repeat until stable.
Gradient boosting	Many small decision trees added together, each fixing the previous ones' mistakes. Excellent for tables of data.
SHAP	A method that explains *why* a model made a particular prediction, feature by feature.
Calibration	When predicted confidence matches reality (80 % circles contain the house 80 % of the time).
Conformal prediction	A recipe that adjusts any model's uncertainty using held-out data so coverage is *guaranteed*.
Coverage / Sharpness	Share of truths inside the predicted region / how small that region is.
P90 error	The error that 90 % of predictions are better than. Shows the "bad tail".
Data leakage	Accidentally letting the model see information it wouldn't have at prediction time, which makes results look better than they really are.
Entity resolution	Figuring out that two differently written records refer to the same real thing.
Stay point	A place where a GPS trail stays within a small radius for a while (the person stopped there).
Pseudo-label	An approximate label made from other data (e.g. consensus of visits) when real labels are scarce.